NeuroSym-BD: Neuro-Symbolic Violence Analysis from Live News Portals

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1. The AI Dilemma in Conflict Monitoring

Real-time violence event monitoring from live news streams presents an acute trade-off:

- **Pure Neural Models (LLMs / GNNs):** High linguistic flexibility, but prone to hallucinations, false logical jumps, and opaque uncalibrated predictions.
- **Pure Symbolic Systems (Ontologies / Rules):** Deterministic and auditable, but brittle against real-world journalistic syntax and noise.

NeuroSym-BD resolves this tension by marrying data-driven neural extraction with strict CAMEO ontological reasoning in BlazeGraph.

2. Bangladesh CAMEO Domain Extension

Standard international CAMEO (Codes 13–20) omits localized South Asian violence archetypes. We formulated 20 specialized OWL Turtle subclasses:

Code	Local Category	Bangla Context
`BD-PV-01`	Political Clashes	Awami League vs BNP
`BD-PV-03`	Mob Lynching	Gonopituni (Mob beating)
`BD-CV-01`	Hartal Violence	Hartal (Picket arson)
`BD-GV-01`	Custodial Torture	Crossfire / Encounter
`BD-LV-01`	Labor Unrest	RMG wage blockades

Linked to W3C PROV-O with 100% graph metadata generation and 98.0% verbatim sentence grounding.

3. System Pipeline & Hybrid Reasoning Core

Daily Star (176 Articles) ⟶ 8-Stage NLP (159 Events) ⟶ BlazeGraph (3,438 Triples) ⟶ Hybrid Engine

Core Neuro-Symbolic Mechanism:

1. **Neural Forecaster:** BiLSTM and Relational GCN (R-GCN) compute probabilities over future CAMEO states $P(Code_{t+1} \mid Code_{1:t})$.
2. **Symbolic Validator:** Evaluates transition legality against formal rules:
   - *Ontological:* Transitions like Threaten (13) → Mass Violence (20) are rejected.
   - *Geographic:* Enforces Bangladesh 8-division and 64-district administrative hierarchy.
   - *Actor Type:* Restricts armed conflict predicates to security or militant groups.
3. **Resolution:** Hard violations trigger fallback; soft violations penalize confidence.

4. Downstream Applications

- **Graph-RAG Q&A:** LLM natural language interrogation grounded directly in verified SPARQL subgraphs.
- **NL2SPARQL API:** Converts natural language questions into formal SPARQL queries.
- **Anomaly Detection:** Poisson & CUSUM statistical spike detection across districts.

5. Experimental Evaluation

Genuine Conflict Benchmarks

Benchmark	Records	Corrob Yield	Concordance
GDELT 2.0 (Direct URLs)	41,554 stories	29.8% (78/262)	**61.5% Code, 100% Macro**
ACLED (2020–2026)	6,691 events	32.1% (42/131)	**78.6% Code, 100% Macro**
NL2SPARQL Benchmark	50 gold queries	—	**52.0% EA**

Ablation Study (Held-Out Test Set, $N = 150$)

Condition	BinAcc	F1@HV	Std-FPR	Expl
Zero-Shot NLI	62.0%	0.467	31.3%	0.0%
Dense Neural	78.7%	0.590	4.0%	0.0%
Symbolic-Only	39.3%	0.503	86.9%	100.0%
Hybrid (Ours)	**85.3%**	**0.732**	**1.0%**	**98.0%**

*Hybrid precision reaches 96.8% ($p = 0.0129 < 0.05$, exact McNemar test); ACLED achieves 100.0% macro-family concordance. Leakage-resistant article-grouped split achieves 84.1% accuracy and 0.683 F1.

6. Takeaways & Future Directions

- **Guaranteed Provenance:** 100% RDF graph metadata and 98.0% verbatim text grounding.
- **Reproducibility:** 28/28 unit tests pass (‘make test‘).
- **Future Work:** Ingesting multilingual Bengali outlets (*Prothom Alo*) using BanglaBERT.

`github.com/ShMazumder/neuro-symbolic-ai-workspace`

LaTeX TikZposter